Shivam Singh

Ph.D. Scholar in Ultracolde quantum gases

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Education

Ph.D. in Physics, Bennett University **2022–Ongoing**

- Thesis: *Different quantum phases in Ultracold quantum gases*
- Advisor: Prof. Ayan Khan

M.Sc. in Physics, National Institute of Technology, Hamirpur **2017–2019**

- Dissertation: *Exigency for the construction of photonic C-NOT gate*
- Advisor: Dr. Arvind Kumar
- CGPA: 7.04

B.Sc. (Honours) in Physics, Ramjas college, University of Delhi **2014–2017**

- Percentage: 66.19

Research Interests

- **Surface Tension in Quantum Liquids:** we have observed significant interest in understanding liquid-like state formation in ultra-cold atomic gases due to the theoretical developments and experimental verification. The fundamental theoretical proposition suggests that the BMF interaction competes with the effective MF interaction to support the clustering of atoms. However, two decades ago a similar exotic phase was predicted theoretically by considering the competition of two-body and three-body interactions. Therfore, in particluar, we have tried to review both the perspectives in quasi one dimensional (Q1D) and one dimensional (1D) geometry through a pathological model.

- **Josephson oscillation to non-linear self trapping regime:** we have studied particle excitation in Bose system in quasi one dimension and also noted the complementary results in one dimension when the particles are trapped in a double well potential and experiences different interaction competitions. Primarily we follow the two-mode model however, we also incorporate the Bogoliubov quasiparticle method for additional insights. We have further extended our analysis to asymmetric double well potential as this tilted topology is key part of atomtronics where it enables the study of coherent quantum transport and the creation of atom-based SQUIDs.

- **Dipolar interaction and fragmentations:** In particular, we have investigated the fragmentation of the quantum fluid due to the competition between two body mean-field (MF), three body (3B), Beyond mean field (BMF) and dipole-dipole interactions (DDI). Specifically, we concentrated on the MF and DDI interaction with non zero BMF interaction. We have observed that the presence of BMF interaction is essential to observe the fragmentation of the fluid. The physical system under consideration is a quasi one dimensional homogeneous condensate. The fragmentation is analyzed in the light of dispersion mechanism. Additionally, we calculate the condensate fraction and superfluid fraction with changing Hamiltonian parameters.

Skills

- **Programming languages:** Mathematica, Fortran, Python, Qiskit

- **Data analysis:** Matplotlib, pandas, numpy

Publications

- S. S. Adusumalli, K. Senapati, **S. Singh** and A. Khan, Quantum liquid in lower dimensions: From the perspective of surface tension, Physics Letters A, **516**, 129638, (2024).

- **S. Singh**, Ibrar, S. Rashid and A. Khan, Josephson Oscillation and Nonlinear Self-Trapping in Quasi-one-dimensional Quantum Liquid (2026), DOI: https://arxiv.org/abs/2602.00674.

- **S. Singh** and A. Khan, Fragmentation of Quantum Fluid in dipolar Bose-Einstein condensate (2026), DOI: https://arxiv.org/abs/2609.20013.

Selected Talks and Conferences

- **Talk** given on "Quantum liquid in lower dimensions: From the perspective of surface tension" in **2nd conference on Quantum Technologies using Ultra-Cold Atoms (QTUCA)** held on 23-25 September, 2024, IISER, Pune.

- **Talk** given on "Fragmentation and Elementary exciations in ultracold systems" in the **Conference on Nonlinear Systems and Dynamics (CNSD 2025)** held in Bharathidasan University, 10-13th March 2025, Tiruchirappalli,Tamil Nadu.

- Present a **Poster** in **International Conference on Complexity and Nonlinear Dynamics in STEM (CNLDS – 2023)**, IIT Hyderabad, 5-7th June 2023.

- Present a **Poster** in **Cold-Atom Based Quantum Simulators and Forthcoming Quantum Technology Applications**, 30th March- 3rd April, ICTS-TIFR, Bengaluru, 2026.

- Quantum Information and Quantum Technology (**QIQT - 2023**), IISER Kolkata, 8-15th June 2023 (Online).

Workshops

- Attend a **workshop** on **Hands-on workshop on Analytical and Numerical Methods for Soliton** held on 5-6th August 2023, TIET, Patiala.

Other Interests

Reading Classic and Modern Classic Literature, listen and play music, global developments in computing technologies, AI frameworks, and research tools, watch sports, Cooking, Watching Movies

For Letters of Reference

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